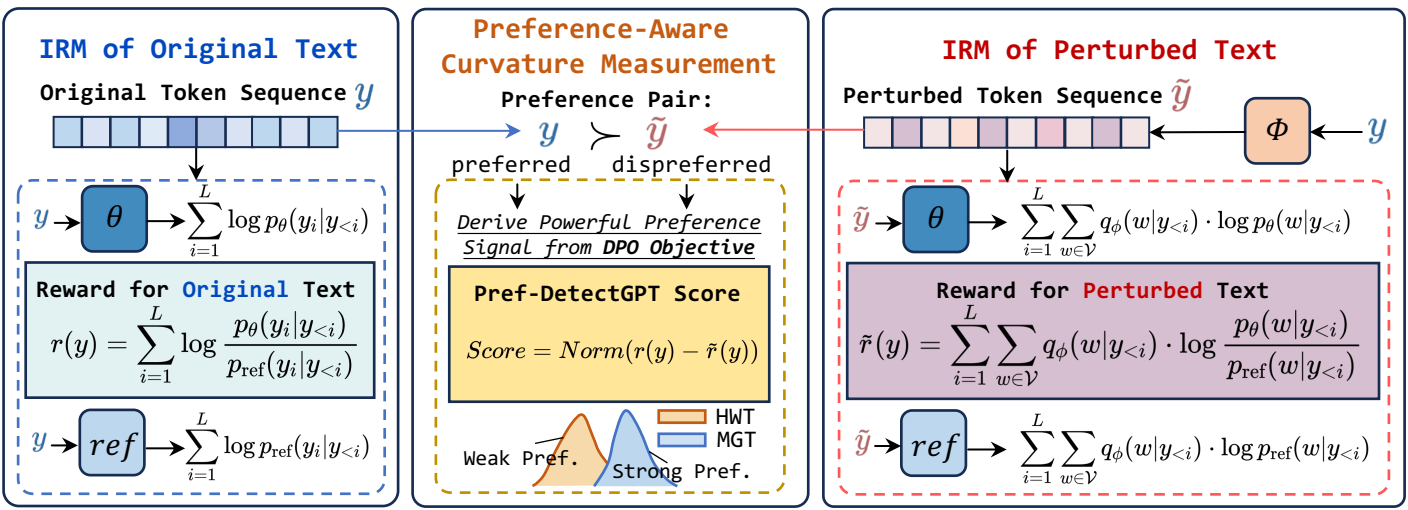

IRM of Original Text
Original Token Sequence y
y → θ → $\sum_{i=1}^{L} \log p_\theta(y_i|y_{<i})$
Reward for Original Text
$r(y) = \sum_{i=1}^{L} \log \frac{p_\theta(y_i|y_{<i})}{p_{\text{ref}}(y_i|y_{<i})}$
y → ref → $\sum_{i=1}^{L} \log p_{\text{ref}}(y_i|y_{<i})$
Preference-Aware Curvature Measurement
Preference Pair:
$y \succ \tilde{y}$
preferred
dispreferred
Derive Powerful Preference Signal from DPO Objective
Pref-DetectGPT Score
$Score = Norm(r(y) - \tilde{r}(y))$
HWT
MGT
Weak Pref.
Strong Pref.
IRM of Perturbed Text
Perturbed Token Sequence $\tilde{y}$
Φ
y
$\tilde{y}$ → θ → $\sum_{i=1}^{L} \sum_{w \in \mathcal{V}} q_\phi(w|y_{<i}) \cdot \log p_\theta(w|y_{<i})$
Reward for Perturbed Text
$\tilde{r}(y) = \sum_{i=1}^{L} \sum_{w \in \mathcal{V}} q_\phi(w|y_{<i}) \cdot \log \frac{p_\theta(w|y_{<i})}{p_{\text{ref}}(w|y_{<i})}$
$\tilde{y}$ → ref → $\sum_{i=1}^{L} \sum_{w \in \mathcal{V}} q_\phi(w|y_{<i}) \cdot \log p_{\text{ref}}(w|y_{<i})$